

Evaluating Retrieval-Augmented Generation Pipelines

A Non-Invasive, Modular Benchmarking Framework for Automated RAG Configuration Analysis

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Problem

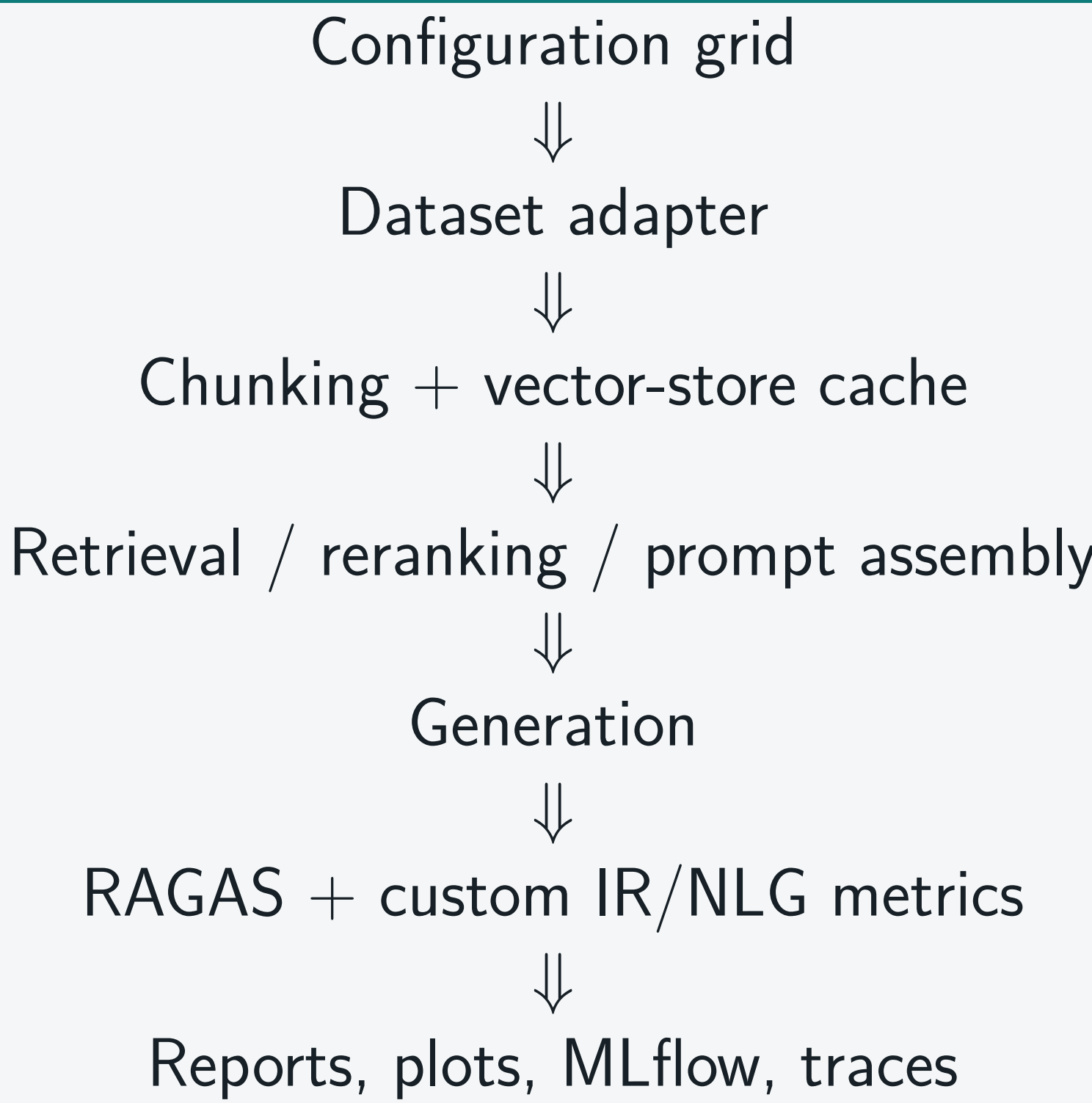
Retrieval-augmented generation (RAG) systems expose a large configuration space. Chunking, overlap, embedding model, vector database, retrieval strategy, reranking, prompt template, generator model, and evaluator choice can all move quality and runtime in different directions.

Research goal: make RAG configuration choices measurable, repeatable, and comparable without rewriting the target pipeline for each experiment.

Framework Contribution

- ▶ Environment-driven benchmark grids for local and OpenAI-compatible model providers.
- ▶ Dataset adapters normalize QA corpora into one benchmark record shape.
- ▶ Reusable retrieval core supports Chroma and LanceDB vector stores, similarity search, MMR, optional HyDE, and cross-encoder reranking.
- ▶ Generation layer captures answer text, time to first token, throughput, and optional GPU statistics.
- ▶ Reporting exports JSON, CSV, Markdown, static plots, MLflow runs, and optional tracing artifacts.

Pipeline



Experiment Space

Dimension	Current values in project
Chunking	recursive, semantic, text, token, markdown
Chunk size	500, 1000, 1500 and variants
Overlap	100, 200, 350 and variants
Retrieval	similarity, MMR
Reranking	none, MiniLM cross-encoder
Embedding	nomic-embed-text:latest snapshot
Datasets	SQuAD / WikiQA-style QA adapters
Models	Qwen3, Qwen3.5, GPT-OSS local runs

Evaluation Lenses

- ▶ **LLM-judged quality:** faithfulness, context recall, semantic similarity, and related RAGAS fields where available.
- ▶ **Retrieval quality:** hit@k, nDCG@k, recall@k, context relevance, with heuristic or gold-document modes.
- ▶ **Answer similarity:** ROUGE-L, BLEU, METEOR, and optional BERTScore precision, recall, F1.
- ▶ **Performance:** total runtime, stage timings, time to first token, tokens per second, and GPU usage when available.

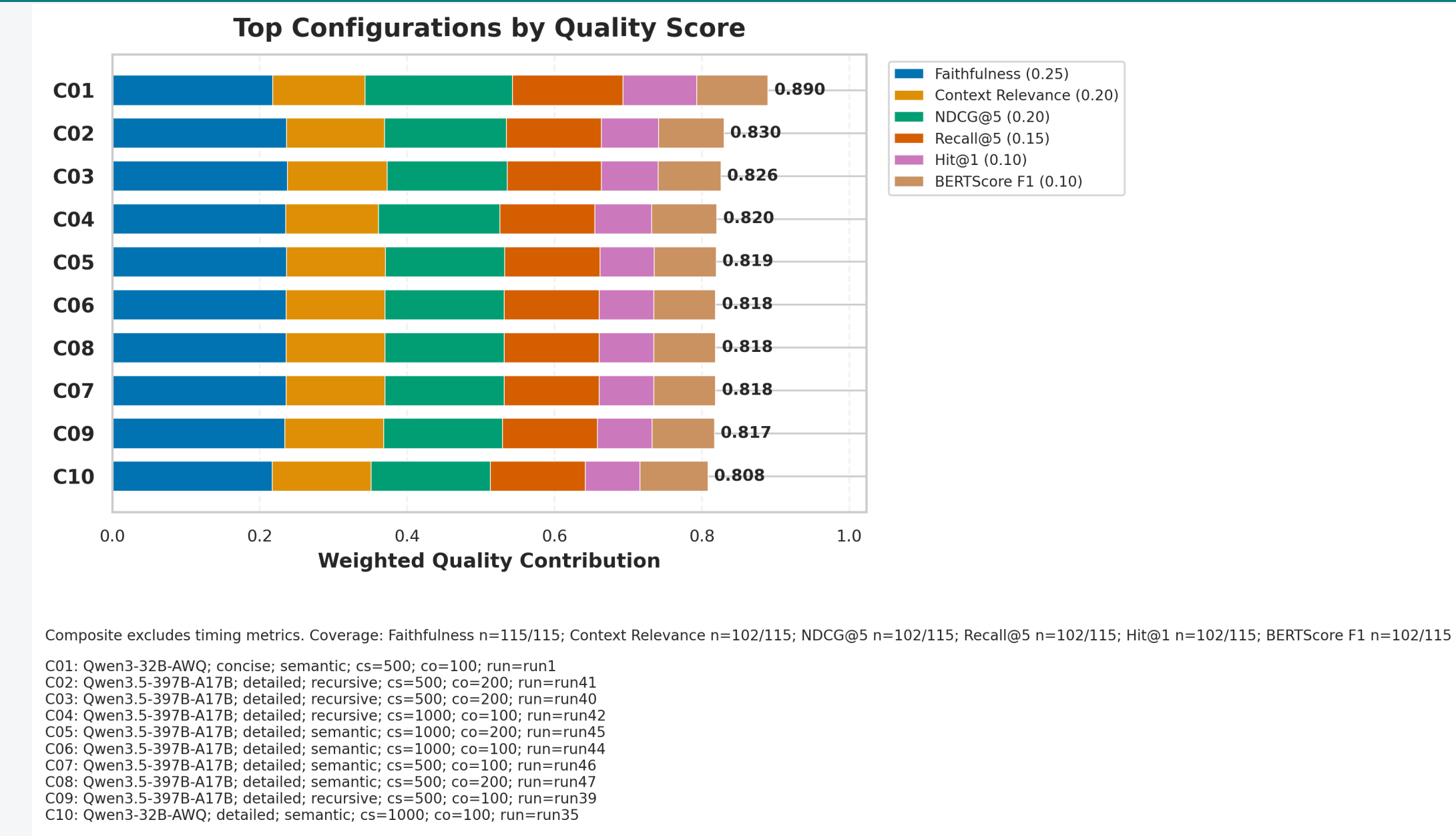
Current Local Results Snapshot

The current repository contains 63 result-summary files and 151 aggregate rows. The manual matrix tracks 128 planned configurations. These figures are useful for poster development, but final submission claims should be regenerated from the final benchmark export.

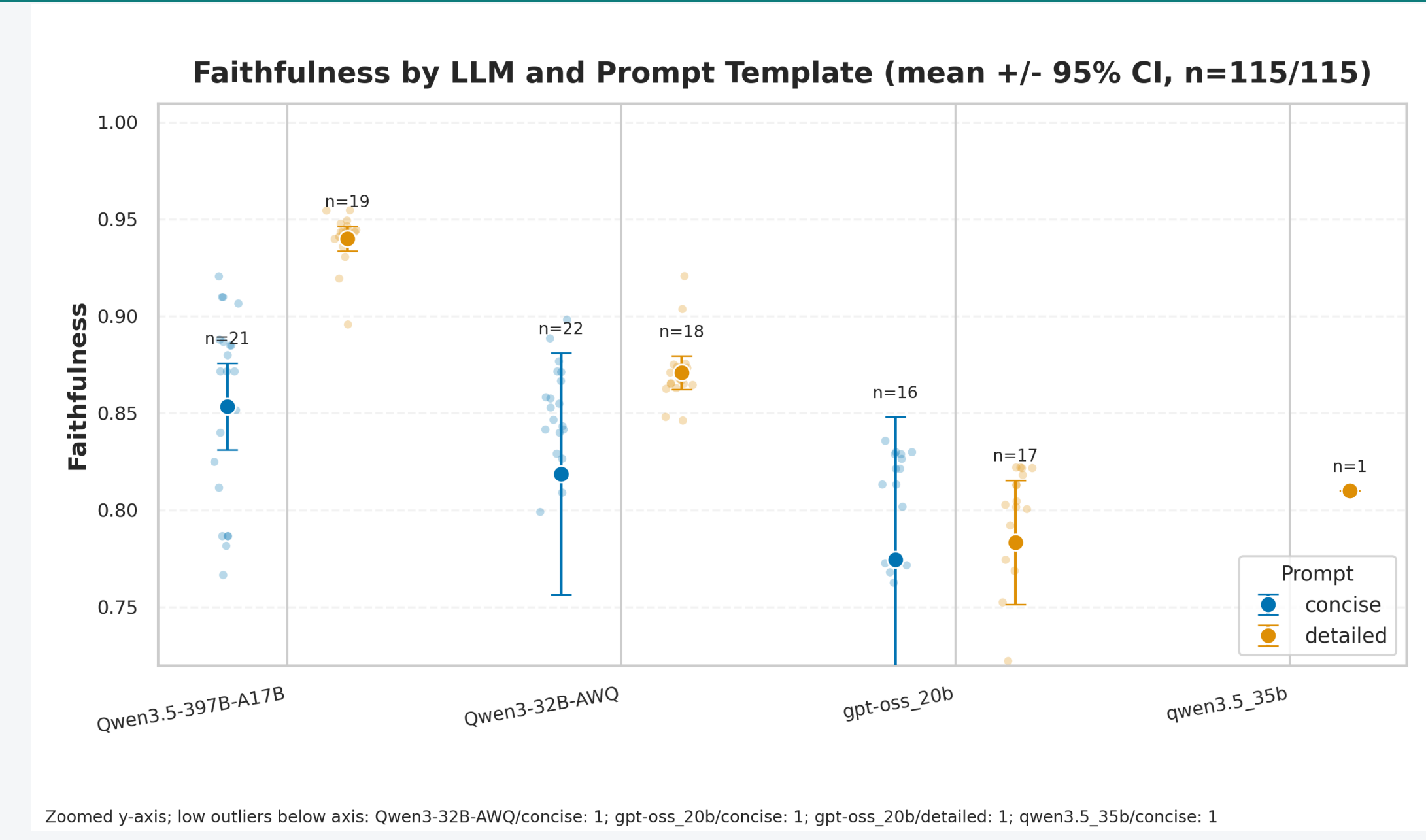
Observed strongest configuration Faith. BERT-F1 nDCG@5			
Qwen3.5-397B-A17B, detailed	0.955	–	–
prompt, recursive chunks			
Qwen/Qwen3-32B-AWQ, concise	0.871	0.969	1.000
prompt, semantic chunks, MMR			
Qwen3.5-397B-A17B, detailed	0.948	–	–
prompt, semantic chunks			

Interpretation: the best run depends on the metric. Strong faithfulness, strong retrieval ranking, and strong lexical/semantic answer similarity do not always select the same configuration.

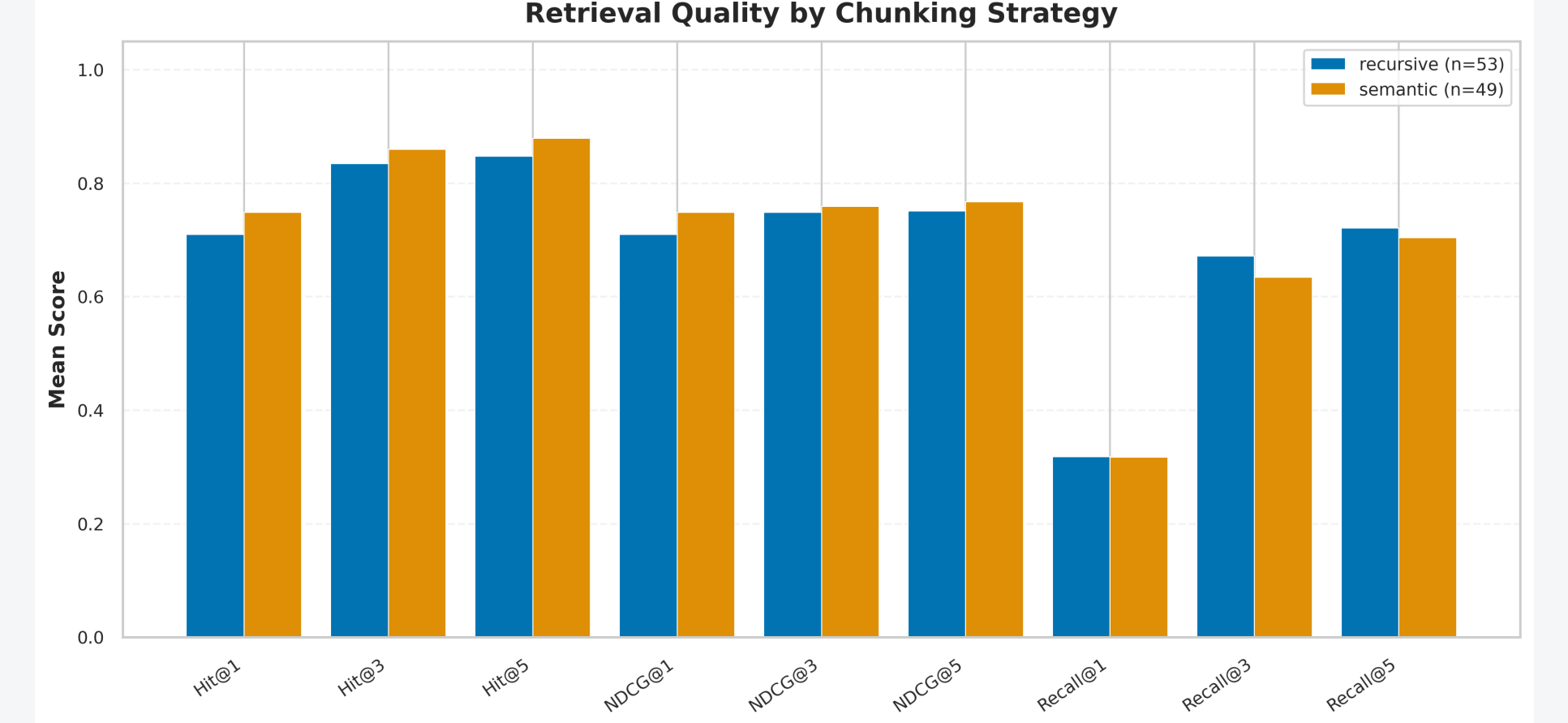
Quality Ranking



Prompt and Model Effects

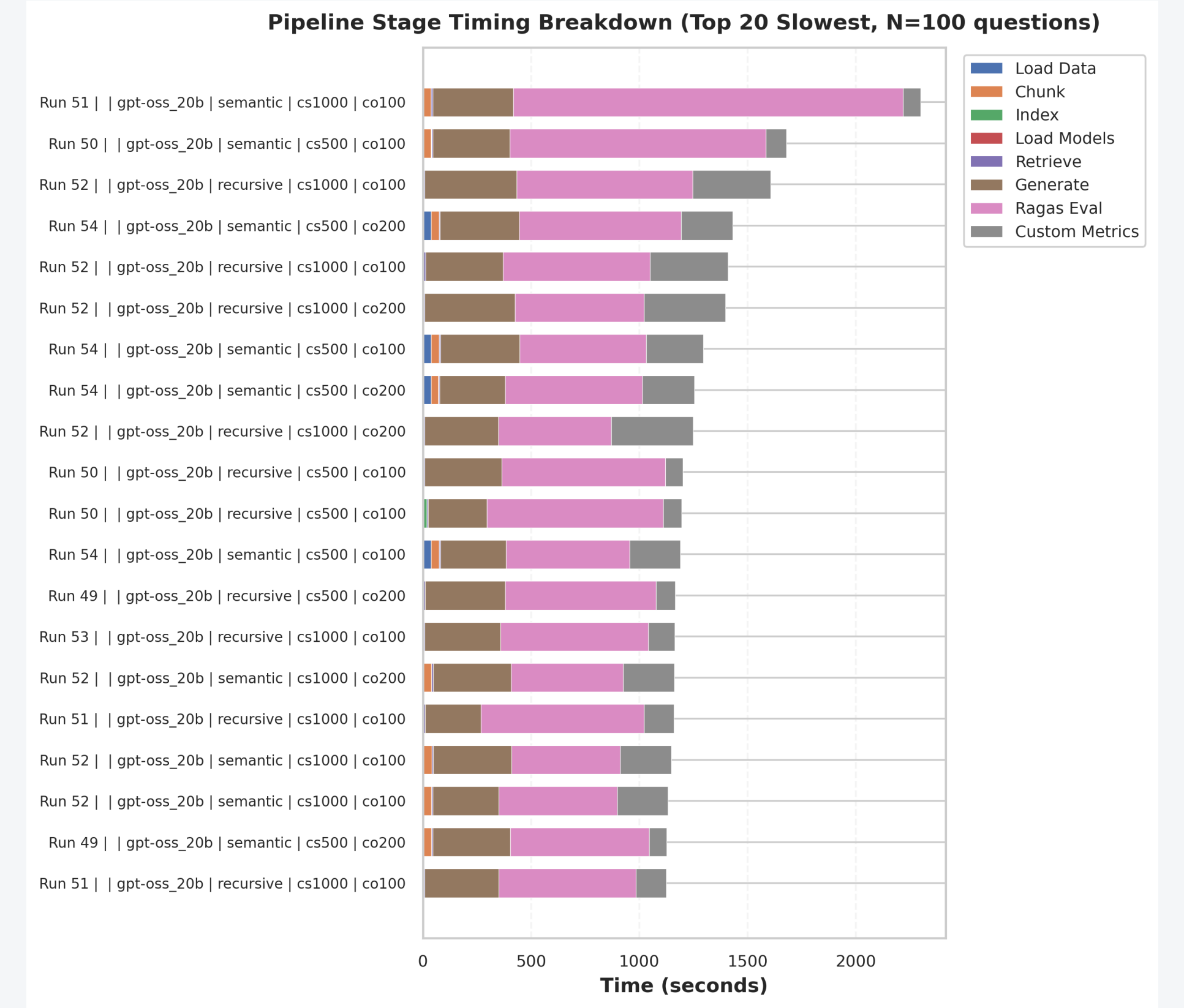


Retrieval Trade-Offs



MMR can improve diversity-sensitive retrieval metrics in some runs, while plain similarity can preserve higher faithfulness in others. Reranking helps in selected settings, but adds latency and should be judged jointly with answer quality.

Runtime and Engineering



- ▶ Persisted vector-store collections avoid unnecessary re-embedding.
- ▶ Stage-level timings separate indexing, retrieval, generation, and evaluation bottlenecks.
- ▶ Sequential execution matches local GPU constraints and reduces resource contention.

Limitations

- ▶ Current results are local, hardware-dependent, and partly duplicated by reruns.
- ▶ SQuAD/WikiQA-style corpora may not transfer to domain RAG workloads.
- ▶ LLM evaluator reliability, stochastic generation, and provider changes remain validity threats.
- ▶ Manual matrix entries must be reconciled with generated result files before publication.

Takeaways for Practitioners

1. Benchmark complete RAG configurations, not isolated components.
2. Report multiple metrics because each metric rewards different behavior.
3. Store machine-readable artifacts for every run.
4. Treat prompt template and retrieval strategy as first-class tuning variables.
5. Keep limitations visible when using LLM-judged metrics.

Next Steps Before Conference

- ▶ **TODO:** insert final conference name, affiliation, contact, QR code.
- ▶ **TODO:** rerun final benchmark export and update table/plots.
- ▶ **TODO:** decide whether poster story favors framework, results, or both.
- ▶ **TODO:** add final paper/repository link.

References: Lewis et al. (2020), Es et al. (2023), Rajpurkar et al. (2016), Lin (2004), Zhang et al. (2020). See references.bib.